

## Stage 1.1 — Liftoff Check & Gravity-Turn Loss (First Principles)

**Era:** Redstone/Mercury-Redstone class, ~1961 (fictional program), continuing Project Arrowhead. **Target time:** ~1.5 hours. **Prerequisite:** Stage 1.0 (Range Table Targeting).

### Why this problem exists

1.0 handed you  $\Delta v$  losses as a given number, the way a real FDO would receive it from the trajectory design group. This problem is that group's work: where the number actually comes from, and — as a bonus you didn't ask for — a check on whether the vehicle spec you were handed even makes sense.

### Part 1 — Sanity-check the vehicle

Before modeling anything about the ascent, check the most basic thing: can this vehicle actually leave the pad? A rocket needs thrust greater than its own weight at ignition, full stop.

**Candidate spec** (proposed by the vehicle design group):  $W_0 = 66,000$  lbf,  $W_p = 40,000$  lbf burned over  $t_b = 143$  s,  $I_{sp} = 235$  s.

$$\begin{aligned}\dot{W} &= W_p / t_b && \text{(weight flow rate)} \\ T &= \dot{W} \cdot I_{sp} && \text{(thrust, from the definition of } I_{sp}) \\ T/W_0 &= \text{liftoff thrust-to-weight ratio}\end{aligned}$$

Compute  $T/W_0$  for the original spec. A real booster of this class typically lifts off around  $T/W_0 \approx 1.2$ ; anything at or below 1.0 cannot lift off at all.

**What you should find:**  $T/W_0 \approx 0.996$ . This vehicle cannot fly. It was never caught because 1.0's calculation only used the *integrated* rocket equation (total  $\Delta v$  from the mass ratio), which doesn't care about instant-by-instant thrust — a completely reasonable thing to miss if you never build a real time-history of the flight, which is exactly what 1.0 didn't do and this problem does.

### Part 2 — Corrected data, and the real ascent

The propulsion group reviewed the spec and traces the error to an overstated liftoff weight. **Corrected value:**  $W_0 = 55,000$  lbf ( $W_p$ ,  $t_b$ , and  $I_{sp}$  unchanged). Recompute  $T/W_0$  and confirm it's now reasonable (target: ~1.2).

With the corrected vehicle, the guidance group ran a preliminary trajectory-shaping study — the kind of repetitive, closely-spaced numerical integration that, even in 1961, nobody did by hand end-to-end. That's the "very limited computer" moment for this problem: below is a coarse extract from that run (7 points instead of the hundreds the real integration would use), given to you the way a trajectory analyst would actually receive it.

**Programmed pitch maneuver:** the guidance program holds the vehicle vertical ( $\gamma = 90^\circ$ ) until  $t_p = 13.0$  s, then flies a prescribed pitch profile that eases toward  $15^\circ$  by burnout. (The exact formula isn't needed for this part — it's given in the Verification step below, when you build your own integrator.)

**Given — trajectory checkpoints** (from the preliminary run):

| t (s) | V (ft/s) | $\gamma$ (°) | h (ft)  |
|-------|----------|--------------|---------|
| 0     | 0.0      | 90.0         | 0       |
| 13    | 98.7     | 90.0         | 604     |
| 39    | 499.8    | 46.53        | 6,485   |
| 65    | 1,282.9  | 28.25        | 19,230  |
| 91    | 2,372.1  | 20.57        | 37,878  |
| 117   | 3,888.0  | 17.34        | 63,427  |
| 143   | 6,284.2  | 15.98        | 100,235 |

**Given — vehicle aerodynamic data** (from wind tunnel testing): effective drag parameter  $C_d \cdot A = 8.0 \text{ ft}^2$ .

**Given — atmosphere model:** isothermal exponential atmosphere (the same simplified model Bate, Mueller & White use for atmospheric-drag perturbation work) —

$$\rho(h) = \rho_0 \cdot \exp(-h/H) \quad \rho_0 = 0.0023769 \text{ slug/ft}^3, H = 23,800 \text{ ft}$$

## Authentic method + reference material

**Drag force:**  $D = \frac{1}{2} \cdot \rho(h) \cdot V^2 \cdot (C_d \cdot A)$  — standard aerodynamic drag equation.

**Loss-rate bookkeeping.** At any instant, gravity is costing you  $g_0 \cdot \sin(\gamma)$  of your rocket's  $\Delta v$ -generating capability, and drag is costing you  $(D/W) \cdot g_0$ . This is genuine physics, not an artifact of this drill — it comes directly from the same equation of motion ( $dV/dt = c \cdot \dot{w}/W - D \cdot g_0/W - g_0 \sin \gamma$ ) that a full trajectory integration solves.

**The hand technique:** compute both loss rates at each of the seven given checkpoints, then integrate (trapezoidal — average the rate at each pair of adjacent checkpoints, multiply by the time between them, sum) to get total gravity loss and total drag loss over the burn. This is *not* the same as integrating the full coupled trajectory yourself — that's a much harder, numerically unstable problem by hand (ask your instructor about the war story where the first attempt at this diverged spectacularly with only five hand-sized steps). Using given trajectory checkpoints and just integrating the loss *rates* sidesteps that entirely, and — surprisingly — still gets you a very good answer, as you'll see in Verification.

## Worksheet

### Part 1 — original spec:

| Quantity                       | Value |
|--------------------------------|-------|
| $\dot{w} = W_p/t_b$            |       |
| $T = \dot{w} \cdot I_{sp}$     |       |
| $T/W_0$ (original, 66,000 lbf) |       |

**Part 2a — corrected spec:**

| Quantity                        | Value |
|---------------------------------|-------|
| $T/W_0$ (corrected, 55,000 lbf) |       |

**Part 2b — loss-rate table.** Fill in for each checkpoint.

|     | t (s)   | V | h       | $\gamma$ | W(t)   | $\rho(h)$ | D | grav. rate = $g_0 \sin \gamma$ | drag rate = $(D/W)g_0$ |
|-----|---------|---|---------|----------|--------|-----------|---|--------------------------------|------------------------|
| 0   | 0.0     |   | 0       |          | 90.0°  |           |   |                                |                        |
| 13  | 98.7    |   | 604     |          | 90.0°  |           |   |                                |                        |
| 39  | 499.8   |   | 6,485   |          | 46.53° |           |   |                                |                        |
| 65  | 1,282.9 |   | 19,230  |          | 28.25° |           |   |                                |                        |
| 91  | 2,372.1 |   | 37,878  |          | 20.57° |           |   |                                |                        |
| 117 | 3,888.0 |   | 63,427  |          | 17.34° |           |   |                                |                        |
| 143 | 6,284.2 |   | 100,235 |          | 15.98° |           |   |                                |                        |

**Part 2c — trapezoidal integration.**

| Interval | $\Delta t$ (s) | Avg grav. rate $\times \Delta t$ | Avg drag rate $\times \Delta t$ |
|----------|----------------|----------------------------------|---------------------------------|
| 0-13     | 13             |                                  |                                 |
| 13-39    | 26             |                                  |                                 |
| 39-65    | 26             |                                  |                                 |

| Interval | $\Delta t$ (s) | Avg grav. rate $\times \Delta t$ | Avg drag rate $\times \Delta t$ |
|----------|----------------|----------------------------------|---------------------------------|
| 65-91    | 26             |                                  |                                 |
| 91-117   | 26             |                                  |                                 |
| 117-143  | 26             |                                  |                                 |

Total gravity loss = \_\_\_\_\_ ft/s Total drag loss = \_\_\_\_\_ ft/s Total  $\Delta v$  losses = \_\_\_\_\_ ft/s

Using  $\Delta v_{\text{ideal}} = 9824$  ft/s (from 1.0's Part A, same vehicle):  $V_{\text{bo}} =$  \_\_\_\_\_ ft/s